

Lab 11: ReactJS – Part 1

Project code structure

```
my-react-app/  
├── src/  
│   ├── assets/  
│   │   ├── profile.jpg  
│   │   └── stem.jpg  
│   ├── components/  
│   │   ├── Footer.jsx  
│   │   └── Navbar.jsx  
│   ├── pages/  
│   │   ├── HomePage.jsx  
│   │   ├── LoginPage.jsx  
│   │   └── Register.jsx  
│   ├── App.css  
│   ├── App.jsx  
│   ├── index.css  
│   └── main.jsx  
├── public/  
├── package.json  
└── vite.config.js
```

1. Use Tailwind framework for CSS Style

1.1 Install Tailwind CSS

Install Tailwind CSS and its Vite plugin.

```
npm install tailwindcss @tailwindcss/vite
```

1.2 Configure Tailwind in `vite.config.js`

Add the Tailwind plugin to the Vite configuration.

```
import { defineConfig } from 'vite'  
import react from '@vitejs/plugin-react'  
import tailwindcss from "@tailwindcss/vite";  
  
export default defineConfig({  
  plugins: [  
    tailwindcss(),  
    react()  
  ]  
})
```

```
    ],  
  })  
}
```

1.3 Import Tailwind in global CSS

Make sure Tailwind is imported in `src/index.css` and `src/App.css`.

```
@import "tailwindcss"
```

2. Create Components

2.1 Create `HomePage.jsx` in `src/pages/`

Build a simple Home page with an image and text.

```
import profilePic from '../assets/profile.jpg';  
import StemBuilding from '../assets/stem.jpeg'  
  
export default function HomePage() {  
  return (  
    <main className=" flex-1 flex-col flex items-center justify-center  
space-y-6">  
  
      <section className=" flex text-start px-4 space-x-4">  
        <img  
          src={profilePic}  
          alt="Profile"  
          className="w-40 h-40 rounded-full shadow-lg border-4 border-  
white mb-6"  
        />  
        <div className='space-y-4'  
          <h1 className="text-4xl md:text-5xl font-bold ">  
            Welcome to My React Site  
          </h1>  
          <p className="text-lg md:text-xl ">  
            Hello! I'm learning <span className="font-semibold text-blue-  
600">React</span>.  
          </p>  
        </div>  
      </section>  
  
      <img  
        src={StemBuilding}  
        alt="StemBuilding"  
        className="max-w-3xl"  
      />  
  
    </main>  
  );  
};
```

```
}
```

2.2 Use `HomePage` in `src/App.jsx`

Render the `HomePage` component inside `App`.

```
import { HomePage } from './pages/HomePage';

function App() {

  return(
    <>
      <HomePage />
    </>
  );
}

export default App;
```

3. Layout Components (Navbar & Footer)

3.1 Create `Navbar.jsx` component in `src/components/`

Build a navigation bar with Home, Register, and Login links.

```
import { Link, useLocation } from "react-router-dom";

export default function Navbar() {
  const location = useLocation();

  const linkClass =
    "px-3 py-2 rounded-md text-sm font-medium hover:bg-gray-100";

  const activeClass =
    "px-3 py-2 rounded-md text-sm font-semibold bg-blue-600 text-white";

  return (
    <header className="bg-white shadow">
      <nav className="container mx-auto flex items-center justify-between px-4 py-3">

        </* Logo */>
        <Link to="/" className="text-xl font-bold text-blue-600">
          MyReactApp
        </Link>

        </* Links */>
```

```

    <div className="flex gap-2">

      <Link
        to="/"
        className={location.pathname === "/" ? activeClass :
linkClass}
      >
        Home
      </Link>

      <Link
        to="/register"
        className={location.pathname === "/register" ? activeClass :
linkClass}
      >
        Register
      </Link>

      <div className="border" />

      <Link
        to="/login"
        className={location.pathname === "/login" ? activeClass :
linkClass}
      >
        Login
      </Link>

    </div>
  </nav>
</header>
);
}

```

3.2 Create **Footer.jsx** component **src/components/**

Add a simple footer with a copyright message.

```

export default function Footer() {
  return (
    <footer>
      <div className="max-w-6xl mx-auto px-4 py-6 flex items-center
justify-center">
        <p className="text-sm ">
          © {new Date().getFullYear()} MyReactSite. All rights reserved.
        </p>
      </div>
    </footer>
  );
}

```

4. Install react-router `npm install react-router-dom`

Add ReactDOM in `src/main.jsx`

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App.jsx";
import "./index.css";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

5. Use Navbar, HomePage, Footer in `App.jsx`

Combine Navbar, HomePage, and Footer in the main layout.

```
import Navbar from './components/Navbar';
import Footer from './components/Footer';
import HomePage from './pages/HomePage';

function App() {
  return (
    <div>
      <Navbar />
      <HomePage />
      <Footer />
    </div>
  );
}

export default App;
```

6. Create pages `LoginPage.jsx` and `RegisterPage.jsx` in `src/pages/`

6.1 Create file `LoginPage.jsx`

```
import { Link } from "react-router-dom";

export default function LoginPage() {
  return (
```

```

    <section className="flex-1 flex items-center justify-center bg-gray-
100 px-4">
      <div className="w-full max-w-md bg-white rounded-2xl border border-
gray-200 p-8 shadow-sm">

        {/* Title */}
        <h2 className="text-3xl font-bold text-center text-gray-800 mb-1">
          Welcome Back
        </h2>
        <p className="text-center text-sm text-gray-500 mb-6">
          Login to your account
        </p>

        {/* Form */}
        <form className="space-y-5">
          {/* Email */}
          <div>
            <label className="block text-sm font-medium text-gray-700 mb-
1">
              Email Address
            </label>
            <input
              type="email"
              placeholder="you@example.com"
              className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm
              placeholder-gray-400
              focus:outline-none focus:ring-2 focus:ring-blue-
600 focus:border-blue-600"
            />
          </div>

          {/* Password */}
          <div>
            <label className="block text-sm font-medium text-gray-700 mb-
1">
              Password
            </label>
            <input
              type="password"
              placeholder="....."
              className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm
              placeholder-gray-400
              focus:outline-none focus:ring-2 focus:ring-blue-
600 focus:border-blue-600"
            />
          </div>

          {/* Submit Button */}
          <button
            type="submit"
            className="w-full rounded-lg bg-blue-600 py-2.5 text-white
font-semibold

```

```

        hover:bg-blue-700 active:scale-[0.99]
        transition-all duration-200"
      >
        Sign In
      </button>
    </form>

    { /* Footer */ }
    <p className="mt-6 text-center text-sm text-gray-600">
      Don't have an account?{" "}
      <Link
        to="/register"
        className="font-medium text-blue-600 hover:underline"
      >
        Register
      </Link>
    </p>
  </div>
</section>
);
}

```

6.1 Create file **RegisterPage.jsx**

```

import { Link } from "react-router-dom";

export default function RegisterPage() {
  return (
    <section className="flex-1 flex items-center justify-center bg-gray-100 px-4">
      <div className="w-full max-w-md bg-white rounded-2xl border border-gray-300 p-8">
        { /* Title */ }
        <h2 className="text-3xl font-bold text-center text-gray-800 mb-2">
          Create Account
        </h2>

        { /* Form */ }
        <form className="space-y-5">
          { /* Name */ }
          <div>
            <label className="block text-sm font-medium text-gray-700 mb-1">
              Full Name
            </label>
            <input
              type="text"
              placeholder="Your name"
              className="w-full rounded-lg border border-gray-300 px-4 py-2.5 text-sm
                focus:outline-none focus:ring-2 focus:ring-blue-

```

```

600 focus:border-blue-600"
    />
  </div>

  {/ * Email */}
  <div>
    <label className="block text-sm font-medium text-gray-700 mb-
1">
      Email Address
    </label>
    <input
      type="email"
      placeholder="you@example.com"
      className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm
                                focus:outline-none focus:ring-2 focus:ring-blue-
600 focus:border-blue-600"
      />
    </div>

    {/ * Password */}
    <div>
      <label className="block text-sm font-medium text-gray-700 mb-
1">
        Password
      </label>
      <input
        type="password"
        placeholder="••••••••"
        className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm
                                focus:outline-none focus:ring-2 focus:ring-blue-
600 focus:border-blue-600"
        />
      </div>

      {/ * Submit Button */}
      <button
        type="submit"
        className="w-full rounded-lg bg-blue-600 py-2.5 text-white
font-semibold
                                hover:bg-blue-700 transition duration-200"
        >
        Create Account
      </button>
    </form>

    {/ * Footer */}
    <p className="mt-6 text-center text-sm text-gray-600">
      Already have an account?{" "}
      <Link
        to="/login"
        className="font-medium text-blue-600 hover:underline"
        >

```



```
        Login
      </Link>
    </p>
  </div>
</section>
);
}
```

7. React Router Setup

7.1 Wrap App with BrowserRouter in src/main.jsx

Enable routing by wrapping your App component.

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App.jsx";
import "./index.css";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

7.2 Define routes in src/App.jsx

Map paths (`/`, `/register`, `/login`) to page components.

```
import { Routes, Route } from "react-router-dom";
import Navbar from "./components/Navbar";
import Footer from "./components/Footer";
import HomePage from "./pages/HomePage.jsx";
import LoginPage from "./pages/LoginPage.jsx";
import RegisterPage from "./pages/RegisterPage.jsx";

function App() {
  return (
    <div className="flex flex-col min-h-screen">
      <Navbar />

      <main className="flex-1 flex flex-col">
        <Routes>
          <Route path="/" element={<HomePage />} />

```

```
        <Route path="/register" element={<RegisterPage />} />
        <Route path="/login" element={<LoginPage />} />
      </Routes>
    </main>

    <Footer />
  </div>
);
}

export default App;
```
